

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>
The file size is 200 GB.

40% of 200 GB is :
$$\left[200 \times \frac{40}{100} = 80 \text{ GB} \right]$$

So, the download takes 80 minutes to reach 40% of the way through the download.

Since Carla can download 2 GB per minute, the time taken to download the first 80 GB is :
$$\left[80 \text{ GB} \div 2 \text{ GB/minute} = 40 \text{ minutes} \right]$$

After reaching 40%, Windows forces a restart to install updates, which takes 20 minutes, and then Carla has to restart the download from the beginning, adding an additional 80 GB to download.

Thus, the time taken to download the remaining 80 GB is :
$$\left[80 \text{ GB} \div 2 \text{ GB/minute} = 40 \text{ minutes} \right]$$

Therefore, the total time to download the file is :
$$\left[40 \text{ minutes} + 20 \text{ minutes} + 40 \text{ minutes} = 100 \text{ minutes} \right]$$
</think>
>
<answer>100</answer>